

- 3 (a)** A curve has equation $y = e^x + \frac{1}{4}e^{-x}$, for $0 \leq x \leq 1$. Find, in terms of π and e , the area of the surface generated when the curve is rotated through 2π radians about the x -axis. [6]

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- (b) Using standard results from the list of formulae (MF19), or otherwise, find the Maclaurin's series for $e^x + \frac{1}{4}e^{-x}$ up to and including the term in x^2 . [2]

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